

Real-World Application ONLINE SHOPPING SYSTEM

A practical relational database design for managing customers, products, orders, payments, shipping and reviews.

SUBMITTED BY	COURSE	SUBJECT
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Assignment Focus

Show how a real-world e-commerce application can be represented using a relational database without adding tables that do not serve a clear business purpose.

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Application & Purpose

What the system does and why a database is needed.

1. Name and Purpose of the Application

Selected application: Online Shopping / E-Commerce Management System

An online shopping system allows customers to browse products, place orders, make payments, track deliveries and submit reviews. The database stores these activities in structured tables so that information remains consistent, searchable and easy to update.

Main problem solved

The database connects customers, products and orders while preserving the history of payments, shipping and reviews. This reduces duplicate data and makes operations such as order tracking, stock checks and sales reporting easier.

Core functions represented in the database

FUNCTION	DATABASE PURPOSE
Customer management	Store customer identity and contact details.
Product catalogue	Maintain products, prices, stock quantities and categories.
Order processing	Record orders and the products included in each order.
Payment tracking	Store the payment method, amount and payment status.
Delivery tracking	Maintain shipment and tracking information.
Customer feedback	Store ratings and review comments for purchased products.

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Required Tables

A focused set of tables for the chosen application.

2. List of Required Tables

#	TABLE	PURPOSE
1	Customers	Customer details and contact information
2	Categories	Product grouping/classification
3	Products	Product catalogue, price and stock
4	Orders	Order header and customer purchase record
5	Order_Items	Products contained in each order
6	Payments	Payment transaction details
7	Shipments	Delivery and tracking details
8	Reviews	Customer ratings and written feedback

Design choice

These eight tables cover the core e-commerce workflow. A separate table is not created for every concept; only entities that need their own stored records are included.

Primary business flow

CUSTOMER

->

ORDER

->

PAYMENT

->

SHIPMENT

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Table Attributes

Columns, keys and the data each table is responsible for.

3. Columns / Attributes for Each Table

Column	Type / Key	Purpose
customer_id	INT, PK	Unique customer identifier
name	VARCHAR(100)	Customer full name
email	VARCHAR(120), UNIQUE	Login/contact email
phone	VARCHAR(15)	Contact number
address	VARCHAR(255)	Delivery address

Column	Type / Key	Purpose
category_id	INT, PK	Unique category identifier
category_name	VARCHAR(80)	Category name, e.g. Electronics
description	VARCHAR(255)	Short category description

Column	Type / Key	Purpose
product_id	INT, PK	Unique product identifier
category_id	INT, FK	Links product to category
name	VARCHAR(150)	Product name
price	DECIMAL(10,2)	Selling price
stock_qty	INT	Available inventory quantity

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Transaction Tables

The tables that record purchases and fulfilment.

Column	Type / Key	Purpose
order_id	INT, PK	Unique order identifier
customer_id	INT, FK	Customer who placed the order
order_date	DATETIME	Date and time of order
status	VARCHAR(30)	Placed / Shipped / Delivered / Cancelled
total_amount	DECIMAL(10,2)	Total order value

Column	Type / Key	Purpose
order_item_id	INT, PK	Unique order-line identifier
order_id	INT, FK	Links line to an order
product_id	INT, FK	Product purchased
quantity	INT	Number of units
unit_price	DECIMAL(10,2)	Price captured at purchase time

Column	Type / Key	Purpose
payment_id	INT, PK	Unique payment identifier
order_id	INT, FK	Order being paid for
payment_method	VARCHAR(30)	UPI / Card / COD etc.
amount	DECIMAL(10,2)	Paid amount
payment_status	VARCHAR(30)	Pending / Paid / Failed / Refunded

Column	Type / Key	Purpose
shipment_id	INT, PK	Unique shipment identifier
order_id	INT, FK	Order sent for delivery
tracking_no	VARCHAR(50), UNIQUE	Courier tracking number
carrier	VARCHAR(60)	Delivery partner
delivery_status	VARCHAR(30)	Packed / Shipped / Out for delivery / Delivered

Column	Type / Key	Purpose
review_id	INT, PK	Unique review identifier
customer_id	INT, FK	Reviewer
product_id	INT, FK	Reviewed product
rating	INT	Rating, e.g. 1-5
review_text	VARCHAR(500)	Written feedback

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Relationships

How the tables connect using primary and foreign keys.

4. Relationships Between the Tables

RELATIONSHIP	CARDINALITY	MEANING
Customers -> Orders	One-to-Many (1:M)	One customer can place many orders. Each order belongs to one customer.
Categories -> Products	One-to-Many (1:M)	One category can contain many products.
Orders -> Order_Items	One-to-Many (1:M)	One order can contain many order items.
Products -> Order_Items	One-to-Many (1:M)	A product can appear in many order lines.
Orders -> Payments	One-to-Many (1:M)	An order can have payment records for payment, retry or refund events.
Orders -> Shipments	One-to-Many (1:M)	An order can be split into multiple shipments when needed.
Customers -> Reviews	One-to-Many (1:M)	A customer can write multiple reviews.
Products -> Reviews	One-to-Many (1:M)	A product can receive many reviews.

Key integrity rule

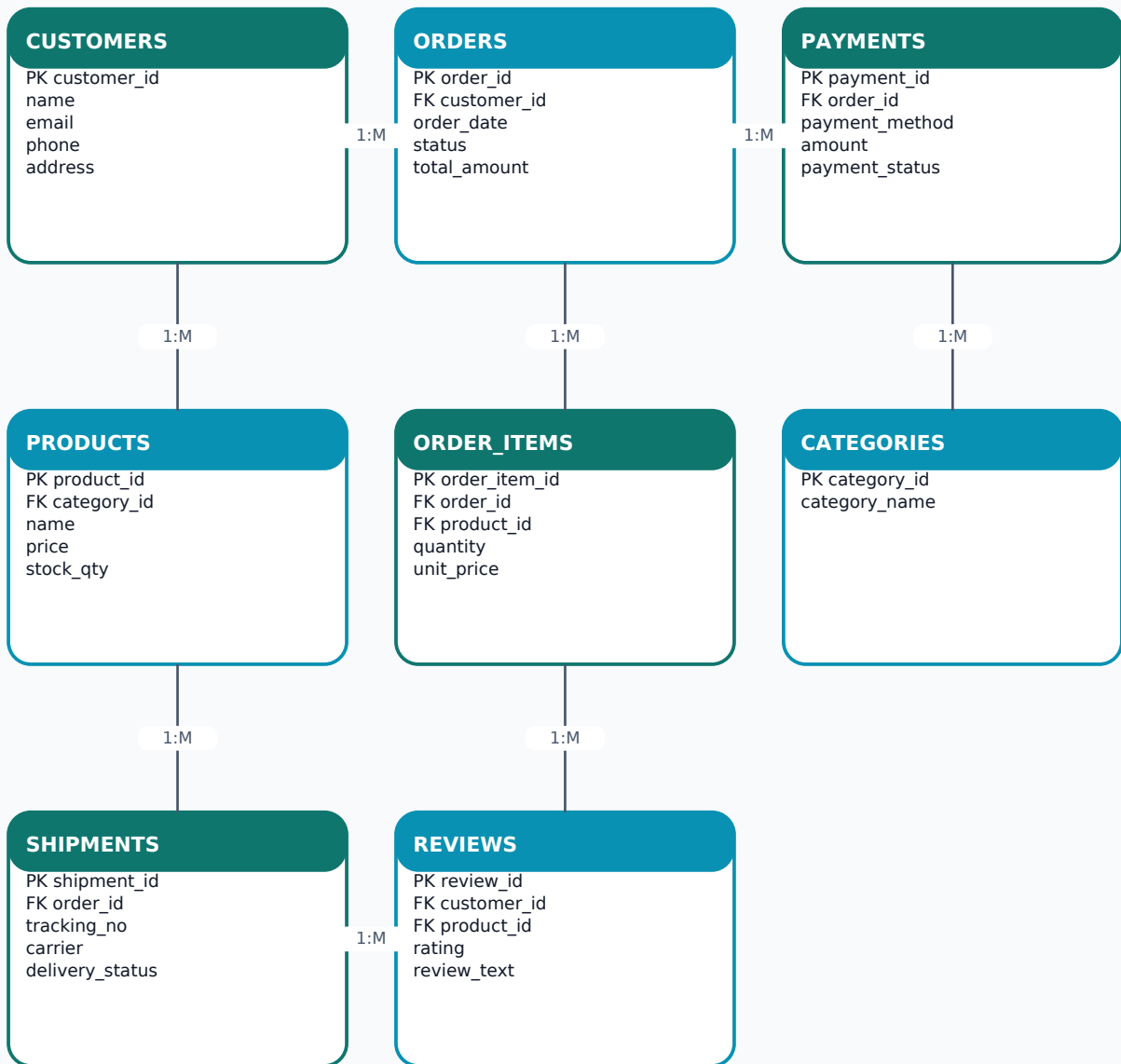
Foreign keys prevent orphan records. For example, an order should reference an existing customer, and an order item should reference both an existing order and an existing product.

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ER-Style Database View

Visual representation of the main entities and their relationships.

Relationship Diagram



PK = Primary Key FK = Foreign Key 1:M = One-to-Many

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Table Explanations

Why each table exists in the application.

5. Brief Explanation of Each Table

TABLE	EXPLANATION
Customers	Stores customer identity and contact information so orders and reviews can be linked to a specific person.
Categories	Groups products into meaningful sections so users can browse and administrators can organise the catalogue.
Products	Stores the items available for sale, including category, price and current stock quantity.
Orders	Represents the purchase transaction made by a customer and stores the order status and total.
Order_Items	Acts as the bridge between orders and products because one order can contain many products and one product can appear in many orders.
Payments	Records how an order was paid and the current payment status.
Shipments	Stores tracking and carrier information used to fulfil and monitor delivery.
Reviews	Stores customer feedback and product ratings for future buyers and product evaluation.

Example transaction flow

#	REAL-WORLD ACTION	TABLES USED
1	Customer 101 logs in and selects a laptop.	Customers + Products
2	The system creates Order 5001 for that customer.	Orders
3	The laptop is added as an order line.	Order_Items
4	Customer pays through UPI.	Payments
5	The order gets a tracking number.	Shipments
6	After delivery, the customer gives 5 stars.	Reviews

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Conclusion

How the database supports the complete application.

Conclusion

The Online Shopping Management System can be effectively represented using eight related relational tables. Primary keys uniquely identify records, while foreign keys connect customers, products, orders, payments, shipments and reviews. This structure keeps the data organised and supports common operations such as placing orders, checking stock, tracking deliveries, processing payments and collecting feedback.

Why this is a good DBMS design

The model is focused on the core business process, avoids unnecessary tables, supports clear relationships, and keeps transaction data separate from master data. The Order_Items table is especially important because it resolves the many-to-many relationship between Orders and Products.

Key DBMS concepts demonstrated

Primary Key	Foreign Key	1:M Relationships	Normalization
Unique record identity	Referential links between tables	One parent, many child records	Reduced duplication and better consistency

END OF ASSIGNMENT